

Hadi Susanto

Professor of Applied Mathematics

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Research Profile

Applied mathematician specialising in nonlinear waves, dynamical systems, and partial differential equations. Substantial international experience in the UK, Europe, and the Middle East, with extensive graduate supervision and leadership in curriculum and programme development. Publications include about 150 journal papers and 30 refereed conference papers, and book chapters, supported by competitive funding from agencies in the UK, UAE, the EU, the Netherlands, and Indonesia.

Selected Achievements

- **Publications:** 120+ journal articles; 30+ refereed conference papers and book chapters.
- **Funding:** PI/Co-PI on projects totalling more than AED 3.5M, plus multiple grants in the UK, EU, and Indonesia.
- **Supervision:** 20+ PhD students (lead and co-supervisor) and several postdoctoral researchers.
- **Editorial Service:** Associate Editor for *IMA Journal of Applied Mathematics* and *Optical and Quantum Electronics*, among others.
- **Invited Talks:** 5–10 invited/keynote/plenary talks per year in Indonesia alone; extensive international speaking record.
- **Recognition:** Faculty Research Excellence Award (Professor Rank) at Khalifa University; national awards and honorary memberships.
- **Community Impact:** Co-founder of SimcovID, a COVID-19 modelling consortium whose tools were adopted by Indonesian national agencies.

Education

PhD (Applied Mathematics)	University of Twente, The Netherlands, Jan. 2006
MSc (Applied Mathematics)	University of Twente, The Netherlands, Jun. 2003
BSc (Mathematics, <i>cum laude</i>)	Institut Teknologi Bandung, Indonesia, Jul. 2001
PGCHE	University of Nottingham, United Kingdom, 2012
Associate Fellow	Higher Education Academy, United Kingdom, Feb. 2013

Academic Appointments

2020–present	Professor, Khalifa University
2019–2022	Professor of Applied Mathematics, University of Essex
2013–2019	Senior Lecturer in Applied Mathematics, University of Essex
2008–2013	Lecturer in Applied Mathematics, University of Nottingham
2006–2008	Visiting Assistant Professor, University of Massachusetts Amherst

Adjunct, Honorary, and Visiting Positions

Adjunct and Honorary Positions

2025–2027 Adjunct Professor, Universitas Islam Negeri Jakarta
2024–2026 Adjunct Professor, Universitas Indonesia
2023–2025 Honorary Professor, Institut Teknologi Sumatera
2018–2020 Adjunct Professor, Institut Teknologi Bandung

Visiting Positions

Institut Teknologi Sepuluh Nopember, Jul.–Aug. 2017, Aug. 2019, Sept.–Dec. 2021.
Institut Teknologi Bandung, Jul.–Aug. 2018.

Administrative Service

2022–2025, Advisory Board Member, Institut Teknologi Bandung.
2022–2025, University Promotion Committee Member.
2022–2023, Department Promotion Committee Member.
2020–2026, Associate Chair for Graduate Studies, Khalifa University.
2020, College of Engineering Undergraduate Curriculum Committee.
2021–2024, Graduate Studies Council Committee Member, Khalifa University.
2016–2019, Director of Research and Impacts, University of Essex.
2016–2019, Ethics Officer, University of Essex.
2016–2019, Chair of Staff–Student Liaison Committee, University of Essex.
2016–2019, Grant Application Officer, University of Essex.
2015, Deputy Graduate Programme Director, University of Essex.
2013–2015, PGT Project Coordinator, University of Essex.
2013–2015, GTA Coordinator, University of Essex.
2011–2013, Programme Director (MSci & BSc Hons) in Mathematical Physics, University of Nottingham.
2008–2010, Postgraduate Student Teacher Mentor, University of Nottingham.
2008–2010, Postgraduate Teaching Support Officer, University of Nottingham.
2008–2010, Staff Development Officer, University of Nottingham.

Recognition and Awards

Faculty Research Excellence Award 2022 (Professor Rank), Khalifa University, May 2023.
Alumni Award 2022, College of Mathematics and Natural Sciences, Institut Teknologi Bandung.
Elected Honorary Member, Physical Society of Uzbekistan, April 2011.
“World Class Professors” Programme Fellow, Ministry of Research, Technology, and Higher Education, Government of the Republic of Indonesia, 2016–2019, 2022.
Ganesa Prize (Best Student), Institut Teknologi Bandung, Aug. 2000.

Editorial Roles

IMA Journal of Applied Mathematics, 2024–present (Associate Editor).
Optical and Quantum Electronics, 2019–2022 (Editor), 2023–2024 (Associate Editor).
Frontiers in Photonics (Nonlinear Optics section), 2021–2022 (Review Editor), 2022–2026 (Associate Editor).
Mathematics (Special Issue “Mathematical Methods in Nonlinear Waves and Dynamical Systems”), 2019–2020.
Communication in Biomathematical Sciences, 2017–present.

Supervision

Postdoctoral Researchers

Rahmi Rusin, 2023.
Roberto Flores, 2022–2025 (joint with Elena Fantino).
Taufik Sutanto, 2022–2023.
Nianqiang Li, 2016–2018 (joint with Michael Adams and Ian Henning).

PhD Students (Completed and Ongoing)

Mia Siti Khumaeroh, 2025–.
Theodore F. Adriano, 2024–.
Defri Ahmad, 2023–.
M. Ismail Yunus, 2022–.
Luthfi Shahab, 2021–2025.
Abrari Noor Hasmi, 2021–2025.
Omar Forrest, 2021–2025.
Zulkarnain, 2020–2024 (Co-supervisor: Chris Antonopoulos).
Awatf Alwiya, 2016–2019 (Lead supervisor: Chris Antonopoulos).
Rahmi Rusin, 2015–2018.
Yuslenita Muda, 2015–2018.
Amal Jasim Mohammed, 2015–2018 (Lead supervisor: Georgi Grahovski).
Junaid Idrees Mustafa, 2015–2018 (Lead supervisor: Georgi Grahovski).
Rudy Kusdiantara, 2014–2017.
Omar Karakchi, 2014–2017.
Tahani Al-Karkhi, 2013–2016 (Co-supervisor: Edward Codling).
Sendy Phang, 2012–2016 (Lead supervisor: Ana Vukovic; Co-supervisors: Trevor Benson & Philip Sewell).
Hui Jiang, 2010–2013 (Co-supervisors: Trevor Benson & Andrew Cliffe).
M. Irfan Qadir, 2009–2012 (Co-supervisor: Paul Matthews).
Amir Ali, 2009–2012 (Co-supervisor: Jonathan Wattis).
Mahdhivan Syafwan, 2008–2012 (Co-supervisor: Steve Cox).
Saeed Ahmad, 2008–2012 (Co-supervisor: Jonathan Wattis).
Xiang Yi, 2008–2010 (Lead supervisor: J. A. D. Wattis).

PhD Examination Committees

Laila Al Sakkaf, United Arab Emirates University, April 2023.
Alan Mohammed Omar, University of Nottingham, December 2019.
Syed Ghufuran, University of Essex, December 2017.
Maharani, University of Essex, July 2015.
Nijamuddin Ali, Visva-Bharati University, July 2015.
Rashid Shakeer, University of Nottingham, July 2013.
Sahabuddin Sarwardi, Visva-Bharati University, June 2013.

Research Grants

Major Projects (Selected)

Research & Innovation Grant – Students (RIG-S), Khalifa University, “DUSTnBLOOM: Algal Bloom and Nutrient and Dust Deposition Effects Under Climate Change in the Arabian Gulf”, January 2024 – December 2025, H. Susanto (PI) and M. R. Al Shehhi [AED 75K].

Research & Innovation Grant – Students (RIG-S), Khalifa University, “Design, Fabrication, and Analysis of Origami-based Metamaterials”, July 2023 – December 2024, T. Y. Kim (PI), W. Cantwell, H. Susanto, and J. Yang [AED 150K].

Competitive Internal Research Awards (CIRA), Khalifa University, “Building Next Generation Orbit Propagation and Analysis Capabilities”, July 2022 – December 2025, E. Fantino (PI) and H. Susanto [AED 2.1M].

Faculty Startup Grant, Khalifa University, “Machine Learning for Dynamical Systems Analysis”, July 2021 – June 2023, H. Susanto (PI) [AED 880K].

Research and COVID-19 Innovation Consortium, Indonesia Ministry of Research and Technology and Ministry of Finance, “Indonesian COVID-19 Mathematical Modelling Simulations and Applications”, September 2020 – September 2021, N. Nuraini (PI), B. W. Lestari, E. R. M. Putri, and H. Susanto [IDR 300M].

EPSRC, SPIN SPACE (Spatially encoded telecoms and quantum technologies using spin-enabled all-optical switching), April 2015 – March 2018, Essex: I. Henning (PI), H. Susanto, M. Adams, N. Balkan; Bristol: R. Oulton (PI), J. Rorison, E. Harbord, total £2M FEC.

Additional Grants

SFC Global Challenges Research Fund, University of Strathclyde COVID-19 Response Fund 2019/20, “Novel Empirical Models for the Effects of Under-Reporting on the Spread of COVID-19 in Indonesia”, April – July 2020, A. Majumdar (PI), N. Nuraini, and H. Susanto [GBP 6,550].

Lorentz Center Workshop Grant on “Nonlinear PDEs on Metric Graphs and Branched Networks”, Leiden, 27–31 August 2018, €23K.

Ministry of Research, Technology, and Higher Education, Government of the Republic of Indonesia, “World Class Professors” Programme, 2016–2019, ca. IDR 150M (~£8K) per year.

University of Essex “Frontrunners”, October 2015 – March 2016, £2.6K.

Program Riset Institut Teknologi Bandung, “Existence and stability of solitons in the theory of $N = 1$ chiral global supersymmetry in 4D”, 2016, PI: Bobby Eka Gunara, IDR 17M (~£0.87K).

British Council – Indonesia Second City Partnership Travel Grant, 2015, £3K.

London Mathematical Society visitor grants, 2011 and 2014, £1K and £1.3K.

University of Nottingham research summer bursaries, June–September 2012 and 2013, £2.2K and £1.5K.

EPSRC Research Development Fund – Bridging the Gaps, “Creating dark and bright solitons through accelerated transport of an impurity in a Bose–Einstein condensate”, 1 January – 30 June 2012, PI: L. Hackermüller, £19K.

University of Nottingham Interdisciplinary High Performance Computing (iHPC), “HPC simulation of coherent structures in waveguide arrays”, 2010–2013, Co-PI: T. M. Benson, £30K.

University of Nottingham Interdisciplinary High Performance Computing (iHPC), “HPC simulations of atom chips: from nanoscience to cosmology”, 2009–2012, PI: T. M. Fromhold, £30K.

Royal Society travel grant, 2009, £400.

Various travel and participation grants (AARMS–CRM, SIAM NW04, NATO ASI 980684, ISEC 2003, etc.).

Career Highlights

Teaching and Pedagogical Leadership

- Extensive undergraduate and postgraduate teaching and course development in mathematics for science and engineering students across several international institutions.
- Leading development of novel and innovative teaching techniques, incorporating new learning technologies such as screencasts (e.g. using Camtasia).
- Delivering training programmes on international education systems to faculty members at various uni-

versities in Indonesia.

- Extensive supervision of graduate students (MSc and PhD).
- Substantial graduate student advisory committee and examination experience in several developing countries, in particular Indonesia and Uzbekistan.
- Curriculum innovation (undergraduate): revision of mathematics and engineering programmes at several universities in Indonesia.
- Curriculum innovation (graduate): co-developer of the postgraduate (MSc and PhD) programme at Khalifa University; first PG director and college representative on the University Graduate Committee.
- Invited speaker/symposium organiser on teaching and learning, particularly active learning strategies, at international and national meetings.
- Extensive capacity building in Indonesia (almost every year), including:
 - Workshop on “PDEs and Dynamical Systems” (3–4 hours each day), University of Indonesia, 8–9 August 2022.
 - Workshop on “Asymptotic Methods and Perturbation Theory” (10 hours in total), Institut Teknologi Kalimantan, Balikpapan, 7–8 August 2017.
 - Mathematics workshop on “Nonlinear Waves” (5 hours each day), Brawijaya University, 22–24 August 2016.
 - Mathematics workshop on “Dynamical Systems in PDEs” (15 sessions of 2 hours each [6 hours/day], with R. Marangell, University of Sydney), University of Indonesia, 10–15 August 2015.

Research Activities

- Research interests: dynamical systems and applied analysis in the study of nonlinear waves.
- Internal and external research funding from national and international agencies in the UK, UAE, the Netherlands, EU, and Indonesia (2008–present).
- Publications: > 120 full papers in peer-reviewed journals and about 30 papers in refereed conference proceedings and book chapters.
- Joint holder of two Intellectual Property Rights (IPR) registered in Indonesia for COVID-19 analysis; the associated application was used by the Ministry of Research and Technology, Republic of Indonesia.
- Regular invited speaker on research in international conferences; typically 5–10 invited/keynote talks per year in Indonesia alone.
- Member of international organising committees for conferences and minisymposia, including:
 - Minisymposium “Coherent structures of nonlinear evolution and lattice equations”, Equadiff 2019, Leiden, July 2019.
 - Lorentz Center workshop on “Nonlinear PDEs on Metric Graphs and Branched Networks”, Leiden, August 2018 (with M. Chirilus-Bruckner, D. Matrasulov, D. Noja, and D. Pelinovsky).
- Reviewer for about 80 scientific journals and various funding agencies.
- Associate Editor of several leading scientific journals.

Administration and Leadership

- Strategic academic leadership across multiple institutions, including oversight of graduate programmes, research activity, and departmental development.
- Responsible for major initiatives in enrolment management, academic advising, and graduate recruitment, contributing to sustained programme growth.
- Led the development and implementation of research strategy as Director of Research and Director of Impact (University of Essex), including internal preparation for the UK REF 2021 process and increasing departmental participation in grant applications.
- Experienced in quality assurance, curriculum governance, and policy development through service on

promotion, graduate studies, curriculum, and ethics committees.

- Contributed to institutional leadership development as a selected participant in the Essex Strategic Leaders programme (intensive senior management training).
- Led departmental and faculty-level strategic reviews, including the Mathematics Area Review at the University of Essex, evaluating trends and shaping future academic priorities.
- Broad administrative engagement beyond academia, including work with schools, community organisations, and professional bodies related to mathematics education.

Community Engagement and Public Impact

- Co-establishing Simcovid (<https://simcovid.github.io/>), a COVID-19 research consortium comprising Indonesian researchers who have constructed models and run simulations related to COVID-19 in Indonesia. The team's "COVID meter" (<https://ppms.itb.ac.id/simcovid/>) was adopted by Kemenristek-BRIN as an alternative measure informing national public policy.
- Involved in the COVID-19 Accelerated Response Task Force (Gugus Tugas Percepatan Penanganan COVID-19) of West Java province, serving as a consultant for the West Java Provincial COVID-19 Task Force.
- Elected as a member of the "World Class Professors" programme (among only around 40 Indonesian academics working abroad) by the Ministry of Research, Technology, and Higher Education, Republic of Indonesia, since 2016. Activities include discussions with university senior management, talks, and workshops aimed at enhancing the quality of human resources in Indonesian universities.
- Actively involved in Indonesian Muslim communities, such as Chair of Nahdlatul Ulama Executive Board – UK Branch (PCINU UK, 2006–2019) and the Indonesian Muslim Community in Abu Dhabi (KMMI Abu Dhabi, 2023–present).

Miscellaneous Professional Activities

- Recognised as one of the prominent figures in Indonesia in COVID-19 analysis; interviewed by national and international television, radio, and newspapers.
- Author of the popular science book "Tuhan Pasti Ahli Matematika" (God Must be an Expert in Maths), published by a major Indonesian press in 2015. The book is sold out and has been reviewed in various newspapers, blogs, YouTube, Facebook, Twitter, etc.; it is rated 3.6/4.0 on Goodreads.com and 4.6/5.0 on Google Store.
- Featured in several national newspapers and magazines for public engagement in mathematics and literature, including:
 - *The Jakarta Post* (21 August 2017), daily English-language newspaper in Indonesia, in an article titled "Simple language needed to make science enjoyable in Indonesia", profiling science communication activities.
 - *Hidayatullah* (December 2017), monthly national Indonesian magazine, six-page interview on family, education, religion, research, mathematics, and literature.
 - *Koran Tempo* (18 May 2008), daily national newspaper, one-page interview on mathematics and literature.
 - Multiple additional news items reporting on activities.
- One of only three invited speakers at the national Science Communication seminar (11 August 2017) on "Building a Science-based Society", organised by the Ministry of Research, Technology, and Higher Education, Indonesia, attended by hundreds of academics and covered by numerous media outlets. The seminar was part of the 22nd commemoration of the Indonesian National Technology Awakening Day (Hari Kebangkitan Teknologi Nasional).
- Active science communicator on social media via a personal Facebook page posting short popular

science and mathematics articles; typical posts receive hundreds of likes and shares. For example, a post on 14 February 2016 on a possible Indonesian autistic savant went viral (about 11K views, c. 400 shares) and was covered by several national newspapers.